

Notes: Alessandria, Kaboski, and Midrigan (AER, 2010)

Inventories, Lumpy Trade, and Large Devaluations

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1 Big picture

- **Question.** How do delivery lags and fixed transaction costs in international trade affect importer inventory behavior, measured trade costs, import dynamics, and exchange-rate pass-through during large devaluations?
- **Setting.** The paper studies importing as an inventory-management problem. Importers face nontrivial shipping/procedural delays, fixed costs of ordering, and uncertain demand, so they order infrequently and carry inventories.
- **Core empirical facts.** Importers hold more inventories than nonimporters; imported orders are larger and less frequent than domestic orders; disaggregated trade flows are extremely lumpy; and large devaluations are followed by import collapses, extensive-margin contractions, and gradual retail price increases.
- **Core model.** A monopolistically competitive importer sells a storable imported good, faces a fixed cost of importing, a one-period delivery lag, and stochastic demand. These features generate an (S, s) -type import rule and inventory-dependent pricing.
- **Main quantitative result.** The measured fixed cost is only about 3.6 percent of the average shipment, but the tariff equivalent of fixed costs plus lags is about 20 percent because firms economize on transaction costs by holding costly inventories.
- **Main dynamic result.** After a devaluation, existing inventories become excessive relative to the new desired stock. Importers stop ordering, run down inventories, reduce markups relative to replacement cost, and only gradually raise retail prices.
- **Economic takeaway.** International trade frictions are not only iceberg costs. Their microstructure matters: fixed ordering costs and time lags convert trade into a lumpy inventory problem, which changes both the level of trade and the short-run response of trade and prices to shocks.

2 Incremental contribution of the paper

- **From iceberg costs to micro trade frictions.** The paper moves beyond the standard iceberg-cost representation of trade barriers by modeling delivery lags and fixed transaction costs explicitly.
- **A bridge between international trade and inventory theory.** The paper imports the logic of (S, s) inventory models into international trade, showing that ordering frictions can rationalize infrequent shipments and high importer inventory holdings.
- **New micro evidence on importer behavior.** The paper combines World Bank trade-cost data, Chilean plant-level inventory data, a transaction-level US steel wholesaler dataset, and disaggregated US export data to document the mechanisms directly.
- **A quantitative reinterpretation of trade costs.** Direct fixed costs look small relative to shipment values, but the full cost of trading internationally is much larger once inventory carrying costs and delivery lags are included.
- **A new mechanism for incomplete pass-through.** The paper shows that prices can respond gradually to exchange-rate shocks even without menu costs or strategic complementarities because firms price off the marginal value of inventories, not simply the replacement cost.

- **A distinct prediction from standard fixed-export-cost models.** In sunk-cost export participation models, short-run trade responses are often smaller than long-run responses. Here, storage plus fixed ordering costs make short-run trade responses larger because firms run down inventories.
- **A devaluation-dynamics contribution.** The paper explains why emerging-market devaluations generate sharp import collapses, large extensive-margin contractions, and slow retail price pass-through.
- **Empirical validation of the inventory mechanism.** The paper tests micro predictions: goods recently imported before a devaluation experience larger post-crisis trade drops, and goods with higher inventories/lumpiness/depreciation have lower short-run pass-through.

3 Data and measurement

3.1 Direct measures of time and fixed trade costs

- The authors use the World Bank Doing Business database to measure the time and monetary costs of importing and exporting containers.
- Costs include document preparation, customs clearing, technical control, and port/terminal handling, but exclude tariffs, international freight, and inland transport.
- Time lags are large: importing procedures typically take several weeks in the countries studied.
- Monetary costs are not proportional to shipment size, so they create economies of scale in ordering.

Interpretation: These frictions imply that importers do not want to import continuously. Instead, they save on fixed ordering costs by importing in large batches and carrying inventories between shipments.

3.2 Inventory evidence from Chilean plants

- The Chilean plant-level data cover a balanced panel of 1,798 manufacturing plants from 1990 to 2001.
- The data record materials inventories, finished-goods inventories, material purchases, sales, and imported materials.
- Importers hold more inventories than nonimporters. Nonimporters hold about 20.6 percent of annual purchases in inventories, while importers hold about 29.8 percent.
- Imported-input intensity is positively related to inventory holdings. Moving from all domestic sourcing to all imported sourcing roughly raises inventory holdings by about two-thirds across specifications.
- In monthly terms, plants hold around 2.1 months of domestic inputs but about 4.3 months of imported goods.

Interpretation: The plant data give a direct target for the model: importing should produce higher inventory-to-purchase ratios than domestic sourcing.

3.3 Transaction-level evidence from a US steel wholesaler

- The transaction-level dataset covers a US steel wholesaler from 1997 to 2006.
- The firm makes both domestic and foreign purchases, allowing within-firm comparison of order size and order frequency.
- Imported orders are about 50 percent larger after controlling for good-year fixed effects and price.
- Imported goods are ordered less frequently: the mean interval is about 151 days for foreign purchases versus 85 days for domestic purchases.

Interpretation: The same firm behaves differently when purchasing foreign versus domestic goods. This makes the inventory mechanism more compelling because it is not only a comparison between different types of firms.

3.4 Lumpiness in disaggregated US exports

- The authors use monthly data on US exports at the HS-10 good level by destination and port.
- The six destination countries are Argentina, Brazil, Korea, Mexico, Russia, and Thailand, each of which experienced a large devaluation.
- Disaggregated trade is very concentrated in a few months. For many countries, the top month accounts for about 40–50 percent of annual trade, and the top three months account for about 70–85 percent.
- This lumpiness provides a calibration target for the model, especially the Herfindahl-Hirschman index of monthly trade concentration.

Interpretation: The export data show that lumpy trade is not just a feature of one steel wholesaler. It is pervasive in highly disaggregated international trade flows.

4 Models and empirical specifications

4.1 Demand and timing

An importer faces CES demand for its variety:

$$q(p, s, v) = \min\{e^v p^{-\theta}, s\},$$

where p is the retail price, v is an idiosyncratic demand shock, θ is the elasticity of demand, and s is beginning-of-period inventory.

The importer has two key frictions:

- a fixed cost f of importing whenever a positive order is placed;
- a delivery lag: goods ordered today arrive next period, so current sales cannot use current imports.

Inventory evolves according to

$$s' = (1 - \delta)[s - q(p, s, v) + i],$$

where i is the new order and δ is the depreciation or inventory carrying-cost parameter.

4.2 Bellman problem

The importer chooses whether to adjust inventories by importing. Let $V^a(s, v)$ denote the value of adjusting and $V^n(s, v)$ the value of not adjusting:

$$V^a(s, v) = \max_{p, i > 0} q(p, s, v)p - \omega i - f + \beta EV(s', v'),$$

$$V^n(s, v) = \max_p q(p, s, v)p + \beta EV(s', v'),$$

with

$$V(s, v) = \max\{V^a(s, v), V^n(s, v)\}.$$

Interpretation: This problem generates an (S, s) inventory rule: the importer waits until inventory falls sufficiently low, then pays the fixed cost and imports a large batch.

4.3 Inventory-dependent pricing

With isoelastic demand, the optimal price is a markup over the marginal value of inventory:

$$p = \frac{\theta}{\theta - 1} V_s(s, v).$$

The marginal value of inventory is not always equal to replacement cost. It depends on inventory holdings:

- when inventories are very low, stockout risk makes the marginal value of inventory high and prices high;
- when the firm is near the adjustment threshold, fixed costs make inventories valuable because they delay the next costly order;
- when inventories are excessive, the marginal value of inventory is below replacement cost, so firms reduce prices to run down stock.

Interpretation: This pricing logic is central to the exchange-rate pass-through result. Retail prices respond gradually because the relevant marginal cost is the shadow value of inventory, not the current replacement cost alone.

4.4 Tariff equivalent

The paper asks what ad valorem tariff would make a frictionless importer equally worse off. The tariff equivalent τ solves

$$V^*(\tau) = EV(0, v),$$

where $V^*(\tau)$ is the value of an importer facing a tariff but no fixed ordering costs or shipping lags.

Interpretation: This exercise converts the welfare cost of inventory-management frictions into a familiar trade-cost metric.

4.5 Micro empirical tests after devaluations

For the quantity mechanism, the authors estimate regressions of the form

$$M_{ijt} = c_{it} + \rho M_{ij,t-1} + \alpha REC_{ijt} + \beta(I_t \times REC_{ijt}) + u_{ijt},$$

where M_{ijt} is import value for good i through port j in year t , REC_{ijt} indicates whether the good was recently imported, and I_t indicates the crisis year.

For the price mechanism, the authors estimate how short-run pass-through depends on depreciation, inventories, and lumpiness:

$$\frac{\Delta p_{it}}{\Delta p_{i,15}} = c_{jt} + a_0 \ln DEP_i + a_1 \ln INV_i + a_2 \ln HH_i + e_{it}.$$

Interpretation: Recent imports proxy for high inventories. Inventory holdings, lumpiness, and depreciation rates proxy for the inventory forces that should affect quantity adjustment and pass-through.

5 Identification and empirical design

5.1 What the design identifies

- The empirical sections identify systematic differences between imported and domestic sourcing behavior and show that these differences are consistent with inventory management under fixed ordering costs and delivery lags.
- The calibration identifies the size of fixed costs and demand uncertainty that rationalize observed lumpiness and inventory holdings.
- The devaluation exercises identify whether the calibrated inventory mechanism can reproduce salient aggregate and micro patterns following large shocks.

Interpretation: The paper is strongest as a quantitative mechanism paper: it documents facts, builds a parsimonious model to match them, and tests additional predictions.

5.2 Key assumptions

- Importers face nonproportional ordering costs and delivery lags.
- Goods are storable and can be sold out of inventories.
- The devaluation shock is modeled as an exogenous increase in the wholesale relative price of imports, higher interest rates, and lower demand.
- Recent shipments proxy for higher inventories in the micro trade data.
- Depreciation rates, inventory ratios, and HH indices proxy for inventory carrying costs, inventory stocks, and lumpiness in the pass-through regressions.

Interpretation: The model abstracts from the causes of devaluations and focuses on propagation. This keeps the mechanism clear but means the paper does not fully explain why the exchange-rate shock occurs.

5.3 Measurement strengths and limitations

- A strength is the multi-dataset approach: Doing Business data, Chilean plant data, US steel transactions, US export microdata, and Argentine price microdata all point in the same direction.

- Another strength is that the model is disciplined by micro moments, not only aggregate trade dynamics.
- A limitation is that inventory holdings are not observed directly in the US export devaluation data; recent importing is used as a proxy.
- Another limitation is that the model is partial equilibrium and takes the terms-of-trade, interest-rate, and demand shocks as exogenous.

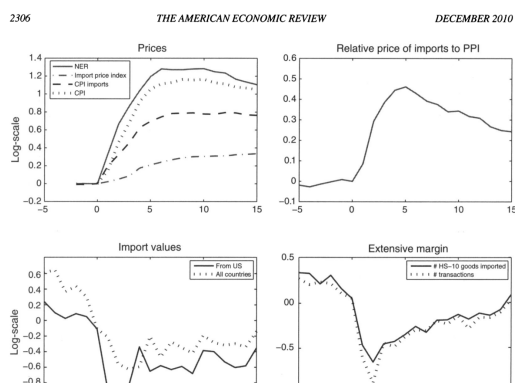
6 Tables and figures

6.1 Figure 1 and Table 1: Devaluation facts and direct trade costs

Read:

- Figure 1 summarizes the Argentine 2002 devaluation: retail import prices rise gradually, the at-the-dock import price rises much faster, import volumes collapse, and the extensive margin contracts.
- Table 1 reports importing lags and fixed procedural costs. Typical importing lags are several weeks, and fixed costs are large relative to median shipment values.
- The table shows that direct non-freight transaction costs can be a sizable fraction of a median shipment, especially for developing countries.

Economic message: The figure motivates the dynamic puzzle, while the table gives the primitive frictions. Importers face both time-to-ship and fixed-ordering problems, so devaluations affect quantities and prices through inventories.



(a) Figure 1: Devaluation in Argentina, 2002

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TABLE 1—TIME AND MONETARY COSTS OF IMPORTING

Country	Number of days	Import cost	US export cost	Median shipment value from the US	Total costs as a fraction of median shipment	Mean shipment value from the US	Total costs as a fraction of mean shipment
Argentina	19	\$1,500	\$625	\$12,300	0.17	\$37,000	0.06
Brazil	23	\$945	\$625	\$13,700	0.11	\$62,200	0.03
Korea	11	\$440	\$625	\$14,500	0.07	\$88,300	0.01
Mexico	23	\$595	\$625	\$10,700	0.11	\$39,300	0.03
Russia	33	\$937	\$625	\$20,700	0.08	\$84,500	0.02
Thailand	20	\$903	\$625	\$12,000	0.13	\$45,700	0.03
Mean							0.03

(b) Table 1: Time and monetary costs of importing

6.2 Tables 2 and 3: Inventory and order-size evidence

Read:

- Table 2 relates Chilean plant inventory holdings to import content. Across specifications, higher import content predicts higher inventories.
- The unweighted specification with employment controls implies that complete domestic sourcing holds around 18 percent of purchases as inventories, while complete importing holds about 36.4 percent.

- Table 3 shows that at the US steel wholesaler, foreign orders are larger and less frequent than domestic orders.
- The foreign order premium remains around 50 percent after controlling for price, and the mean interval between foreign orders is 151 days versus 85 days for domestic orders.

Economic message: Importing changes inventory behavior both across plants and within a single firm. Imported goods are ordered in bigger, less frequent batches, just as a fixed-cost inventory model predicts.

TABLE 2—REGRESSION RESULTS OF INVENTORY HOLDINGS ON IMPORT CONTENT (1990 to 2001)

	Unweighted		Weighted		Robust		Fixed	
	$\hat{\rho}^m$	c	$\hat{\rho}^m$	c	$\hat{\rho}^m$	c	$\hat{\rho}^m$	c
Inventory	0.196 (17.6)	0.213 (75.6)	0.103 (20.1)	0.157 (83.1)	0.219 (49.6)	0.133 (119.0)	0.074 (6.3)	0.226 (84.4)
Materials inventory	0.139 (14.9)	0.16 (67.6)	0.081 (20.2)	0.106 (72.0)	0.154 (50.3)	0.087 (113.4)	0.051 (5.0)	0.169 (73.1)
Finished inventory	0.057 (12.9)	0.053 (47.6)	0.022 (9.6)	0.051 (60.5)	0.041 (51.5)	0.012 (58.2)	0.023 (5.0)	0.057 (54.2)

(a) Table 2: Inventory holdings and import content

TABLE 3—STATISTICS ON LUMPINESS AT A US STEEL WHOLESALER

Panel A: Summary statistics			Domestic		Foreign	
Goods	Purchases	Value	Purchases	Value		
3,321	12,472	\$134 mln	5,634	\$87.8 mln		
Panel B: Premium on imported goods (good-yr fixed effects)						
Amount	Weight	Price				
0.462	0.583	-0.063				
(17.7)	(21.9)	(17.2)				
Panel C: Import size premium controlling for price (good-yr fixed effects)						

(b) Table 3: Lumpiness at a US steel wholesaler

6.3 Table 4 and Figure 2: Lumpy exports and optimal importing

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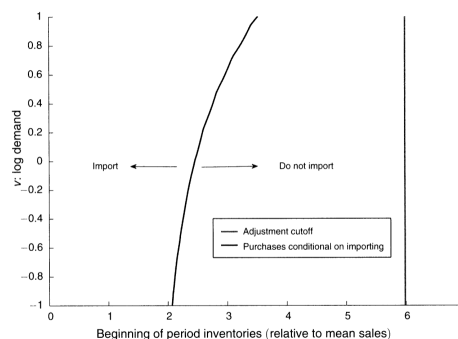
- Table 4 shows that disaggregated US exports to the six devaluation countries are highly concentrated in a few months.
- Except for Mexico, the median good is exported in only a subset of months; the top three months often account for most annual trade.
- Figure 2 shows the optimal import rule: firms import when inventories are low relative to demand and do not import when inventories are high enough.
- Conditional on importing, firms jump back to a high inventory target of several months of sales.

Economic message: The empirical lumpiness is the aggregate trace of firm-level (S, s) ordering. Fixed costs make firms wait, then place large orders.

TABLE 4—LUMPINESS STATISTICS OF DISAGGREGATE US EXPORTS TO DIFFERENT DESTINATION COUNTRIES

	Argentina	Brazil	Korea	Mexico	Russia	Thailand
Number of transactions (in months with trade)	2.2	3.0	4.8	32.3	2.7	3.2
Fraction of months good exported	0.27	0.33	0.40	0.81	0.13	0.26
Fraction of months in year good exported	0.47	0.55	0.70	0.90	0.43	0.55
Herfindahl-Hirschman index	0.40	0.37	0.28	0.21	0.45	0.35
Fraction of ann. trade in top month	0.50	0.47	0.38	0.27	0.53	0.45
Fraction of ann. trade in top 3 months	0.83	0.78	0.70	0.53	0.85	0.79

(a) Table 4: Lumpiness statistics for US exports



(b) Figure 2: Optimal import rules

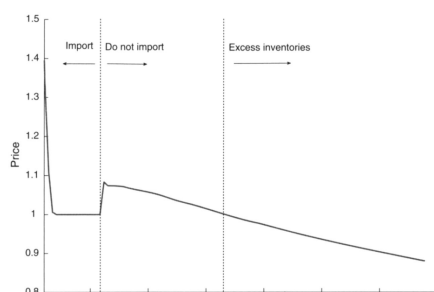
6.4 Figure 3 and Table 5: Pricing rules and calibration

Read:

- Figure 3 shows that retail prices vary with inventory holdings. Prices are high at very low inventory because of stockout risk, elevated near the no-adjustment region because of fixed costs, and low when inventories are excessive.
- Table 5 reports the calibration. The benchmark model matches an HH ratio of 0.44 and an inventory-to-annual-purchases ratio of 0.36.
- The fixed cost is calibrated to 9.5 percent of median monthly revenue, or about 3.6 percent of the mean shipment.
- The tariff equivalent of fixed costs plus lags is about 20 percent, much larger than the measured fixed cost itself.

Economic message: The calibration turns small visible transaction costs into large effective trade barriers because firms respond by carrying costly inventories.

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(a) Figure 3: Optimal price functions

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	Data	Benchmark	Domestic	No fixed cost	No lag	High depreciation
<i>Panel A. Moments</i>						
Used for calibration						
Herfindahl-Hirschmann ratio	0.44	0.44	0.23	0.14	0.32	0.33
Inventory to annual purchases ratio	0.36	0.36	0.21	0.29	0.12	0.25
<i>Additional implications</i>						
Tariff equivalent of frictions	—	0.20	0.09	0.13	0.08	0.32
f (relative to mean shipment)	—	0.036	0.017	0	0.050	0.047
<i>Panel B. Parameters</i>						
<i>Calibrated</i>						
f (fixed cost, rel. median revenue)		0.095	0.025	0	0.095	0.095
Standard deviation of demand, σ		1.15	1.15	1.15	1.15	1.15
<i>Assigned</i>						
Period length		1 month	½ month	1 month	1 month	1 month
Shipping lag		1 month	½ month	1 month	0 months	1 month
Elasticity of demand for imports, θ		1.5	1.5	1.5	1.5	1.5
Elasticity of subs. across imported goods		—	—	—	—	—
Monthly discount factor, β		0.995	0.995	0.995	0.995	0.995

(b) Table 5: Moments and parameters

6.5 Table 6 and Figure 4: Devaluation stylized facts

Read:

- Table 6 reports large terms-of-trade increases and sharp import drops in Argentina, Brazil, Korea, Mexico, Thailand, and Russia.
- The extensive margin explains a large part of the import collapse, especially when measured using transactions or HS-10 goods.
- Figure 4 shows that the short-run import-price elasticity is large, especially in countries where the terms-of-trade shock is sudden.
- These patterns differ from the standard J-curve view in which trade responds slowly at first.

Economic message: Emerging-market devaluations generate fast quantity adjustment, not slow quantity adjustment. This is exactly the kind of pattern an inventory-rundown mechanism can produce.

TABLE 6—SALIENT FEATURES OF LARGE DEVALUATIONS

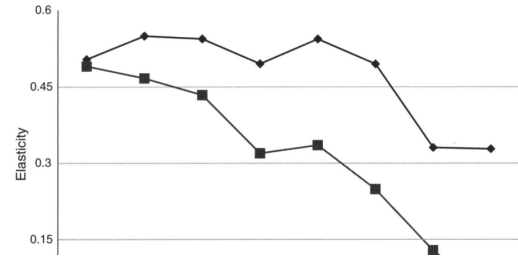
max Δ (PI/PPI)	Trade drop	Extensive margin					
		Method 1		Method 2		Method 3	
		# cards	# goods	# cards	# goods	# cards	# goods
<i>Panel A. Relative price of imports, trade drop, and share due to extensive margin in month with worst trade drop</i>							
Argentina	0.47	-0.76	1.09	0.82	1.25	0.59	1.32
Brazil	0.49	-0.52	0.76	0.46	1.03	0.29	1.04
Korea	0.34	-0.65	0.91	0.68	0.58	0.24	0.75
Mexico	0.55	-0.22	1.34	1.02	1.71	0.46	1.70
Thailand	0.50	-0.59	0.66	0.56	0.31	0.06	0.54
Russia	1.01	-1.83	0.58	0.40	0.93	0.37	1.08
Average	0.56	-0.76	0.89	0.66	0.97	0.33	1.07
	Trade drop	Extensive margin					
		Method 1		Method 2		Method 3	
		# cards	# goods	# cards	# goods	# cards	# goods
<i>Panel B. Average trade drop and share due to extensive margin in three-month window around worst month</i>							
Argentina	-0.69	0.88	0.74	0.98	0.52	1.06	0.62
Brazil	-0.45	0.54	0.36	0.48	0.20	0.49	0.29
Korea	-0.51	0.94	0.70	0.75	0.21	0.87	0.31
Mexico	-0.13	1.61	1.47	2.62	0.91	2.61	0.90
Thailand	-0.53	0.75	0.68	0.29	0.06	0.52	0.32
Russia	-1.71	0.63	0.45	0.99	0.45	1.17	0.62

(a) Table 6: Salient features of large devaluations

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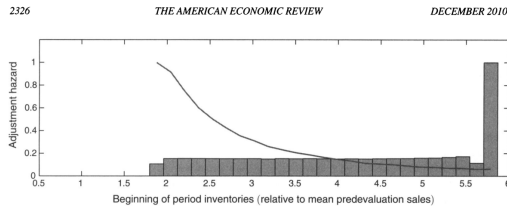
(b) Figure 4: Median import price elasticity

6.6 Figures 5A and 5B: Inventory distributions before and after devaluation

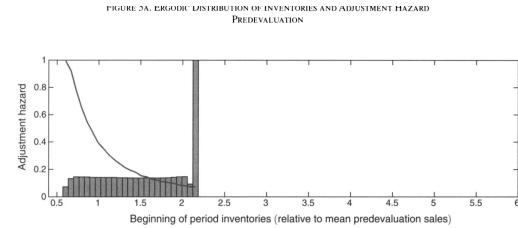
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- Figure 5A shows the predevaluation distribution of inventory holdings and the adjustment hazard.
- Figure 5B shows the postdevaluation distribution. The higher import price lowers desired inventory holdings and shifts the adjustment hazard left.
- After a devaluation, many firms that would have imported before now prefer not to import because they already hold too much inventory relative to the new target.

Economic message: The extensive-margin collapse arises from an endogenous pause in ordering. Firms do not disappear from retail sales; they keep selling out of existing inventories but stop importing temporarily.



(a) Figure 5A: Predevaluation inventories and hazard



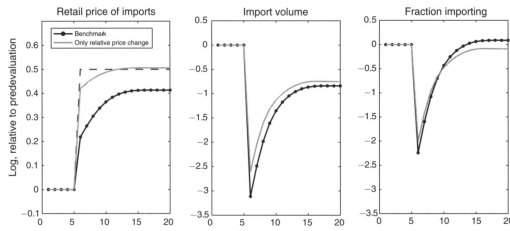
(b) Figure 5B: Postdevaluation inventories and hazard

6.7 Figures 6 and 7: Model response and predevaluation demand drop

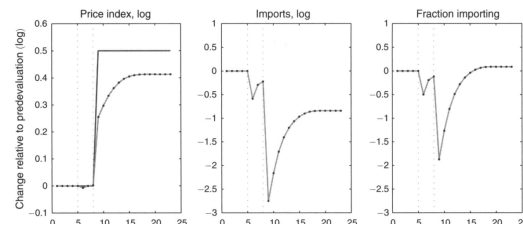
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- Figure 6 shows the model response to a devaluation. Retail import prices increase gradually, import volumes collapse sharply, and the fraction importing drops sharply.
- Immediate pass-through is incomplete because firms lower markups relative to replacement cost while running down inventories.
- Figure 7 shows that if consumption falls before the devaluation, the qualitative devaluation dynamics remain similar.

Economic message: The model produces a trade implosion without needing all firms to suddenly lose access to trade. It is enough that their desired inventories fall sharply.



(a) Figure 6: Response of model economy to devaluation



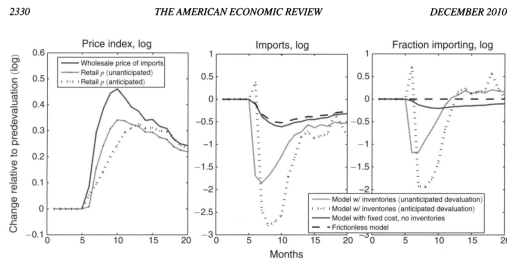
(b) Figure 7: Consumption drops before devaluation

6.8 Figures 8 and 9: Gradual devaluation and recent imports

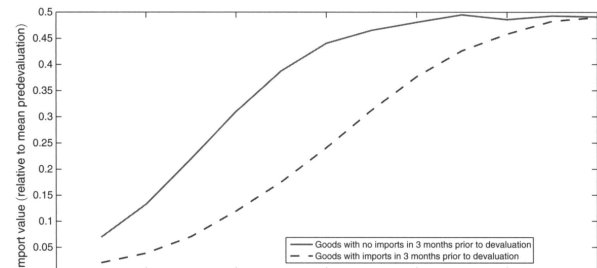
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- Figure 8 feeds the Argentine relative-price path into the model. Even with a gradual devaluation, inventories generate a rapid trade collapse and slow price response.
- The model without inventories would predict a much more gradual decline in imports.
- Figure 9 shows the model prediction that firms with recent predevaluation imports, and thus high inventories, import less after the devaluation.
- This prediction is later taken to the data using recent shipments by good and port.

Economic message: Inventory positions before the shock determine the short-run trade response. Recent importers have more stock to run down, so they leave the import market for longer.



(a) Figure 8: Response to gradual devaluation



(b) Figure 9: Imports by date of last predevaluation import

6.9 Table 7 and Figure 10: Micro evidence for quantity and price mechanisms

Read:

- Table 7 estimates the effect of recent shipments on crisis-year trade values. The interaction of recent shipment with the crisis year is negative and significant for all six countries.
- This means goods imported shortly before the devaluation experience larger post-crisis import declines, consistent with higher inventories being run down.
- Figure 10 illustrates price pass-through predictions. Higher initial inventories and higher depreciation/carrying costs reduce short-run pass-through.

- The price mechanism works because excess inventories lower the marginal value of replacement goods relative to replacement cost.

Economic message: The paper goes beyond fitting aggregate facts. It tests the cross-sectional implication that inventory-rich goods should cut imports more and pass through cost shocks more slowly.

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TABLE 7—EFFECT OF RECENT SHIPMENTS ON VALUE TRADED IN THE CRISIS YEARS

Regressors	Country					
	Argentina	Brazil	Korea	Mexico	Russia	Thailand
Recent shipment (α)	2.46 (20.7)	0.77 (9.3)	1.99 (16.0)	2.14 (21.8)	3.71 (15.1)	1.63 (11.5)
Recent shipment before crisis (β)	-1.51 (-6.1)	-0.89 (-5.3)	-2.68 (-11.14)	-0.60 (-3.1)	-4.04 (-8.9)	-0.57 (-2.1)
Lagged imports (ρ)	0.534 (196.6)	0.969 (511.5)	0.789 (354.5)	0.730 (344.4)	0.393 (100.1)	0.763 (327.1)
R^2	0.41	0.55	0.51	0.42	0.28	0.75

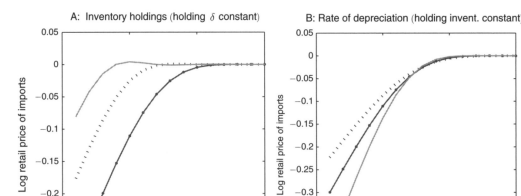
Notes: T -stats in parentheses. The dependent variable is value imported _{it} , where i = commodity, j = exit port, t = year relative to devaluation. Here we annualize the data for value of trade and positive shipments (constructing years around the crisis date rather than calendar years). Using five years of data, we regress value traded on the amount of trade in the last three months of the year (i.e., preceding the crisis) interacted with the year. We control for commodity-year fixed effects, and therefore use variation in port to identify the effect. We also control for the value traded in the previous year (to control for differential levels and trends among ports). The omitted year is the year before the crisis; the interaction effect in year three is the impact in the 12 months following the devaluation.

(a) Table 7: Recent shipments and crisis-year trade

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(b) Figure 10: Inventory holdings, depreciation, and pass-through

6.10 Figure 11 and Table 8: Argentine pass-through evidence

Read:

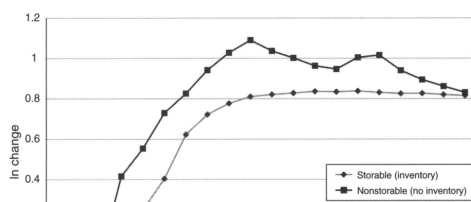
- Figure 11 compares Argentine price dynamics for storable and nonstorable imported goods.
- Nonstorable goods, which have little inventory, follow the nominal exchange rate more closely and pass through faster.
- Table 8 shows that depreciation, lumpiness, and inventory measures are negatively related to short-run pass-through at three months.
- These effects fade by nine months, consistent with a transitory inventory channel.

Economic message: Inventory management slows retail pass-through, especially for goods with high inventories, lumpy ordering, or high carrying costs. The mechanism matters most in the first few months after the devaluation.

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(a) Figure 11: Argentine price dynamics by inventory

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TABLE 8—SHORT-RUN PASS-THROUGH ON DEPRECIATION, LUMPINESS, AND INVENTORIES

	Months after crisis		
	3 months	6 months	9 months
Depreciation	-0.044 (3.1)	-0.026 (1.5)	-0.01 (0.7)
Lumpiness (HH)	-0.124 (-3.4)	-0.081 (-1.8)	-0.001 (-0.0)
Inventory	-0.092 (-3.0)	-0.001 (-0.0)	0.014 (0.5)
Constant	0.006 (0.1)	0.73 (6.3)	0.97 (10.0)
Observations	517	517	517
R^2	0.243	0.142	0.008

Notes: T -stats in parentheses. All regressions include dummies for tradability (imported, with some import share, exportable, etc.); coefficients are not reported. All variables are in logs. The dependent variable is the change in log price over t ($t = 3, 6, \text{ or } 9$) months from the devaluation over the long-run (15-month) price change.

(b) Table 8: Short-run pass-through regressions

6.11 Table A: Labor share in Chilean plants

Read:

- Table A reports labor shares for Chilean plants and helps discipline the local labor share used in the devaluation experiment.

- The model sets the local labor share to 25 percent, consistent with the Chilean evidence and the idea that retail/import distribution uses local inputs.

Economic message: The local labor component moderates how much the wholesale import price shock raises marginal cost at the retail level.

TABLE A. LABOR SHARE IN CHILEAN PLANTS

<i>Panel A. Mean labor share</i>		
	Unweighted	Weighted
Importers	0.211	0.137
Total	0.232	0.136
<i>Panel B. Controlling for import content and log employment</i>		
Constant	−0.043*	0.042*

Table A: Labor share in Chilean plants

7 Short summary

- The paper studies how fixed ordering costs and delivery lags shape international trade.
- Importers hold more inventories than nonimporters, and imported orders are larger and less frequent than domestic orders.
- Disaggregated trade is highly lumpy: annual trade is concentrated in a few months.
- The model is a partial-equilibrium inventory-management problem with fixed ordering costs, one-period delivery lags, and stochastic demand.
- The model matches observed inventory holdings and trade lumpiness.
- The measured fixed cost is about 3.6 percent of a mean shipment, but the tariff equivalent of the frictions is about 20 percent because inventory carrying costs are large.
- After a devaluation, importers hold excess inventories, stop importing temporarily, and reduce markups relative to replacement cost.
- This generates an import collapse, a sharp drop in the extensive margin, and gradual retail pass-through.
- Micro evidence supports the mechanism: recent predevaluation imports predict lower crisis-year trade, and inventory/lumpiness/depreciation predict slower pass-through.

8 Conclusion

8.1 Core conclusion

Alessandria, Kaboski, and Midrigan show that international trade frictions have an inventory-management structure. Delivery lags and fixed costs make trade lumpy, induce importers to hold large inventories, and make measured trade costs understate true effective trade barriers.

8.2 Economic intuition

The core intuition is that fixed costs make importers wait and order in batches, while shipping lags make inventories valuable as a buffer against stockouts. During normal times, firms carry

inventories to economize on ordering costs and avoid being unable to sell. During devaluations, the desired inventory stock falls sharply. Existing inventories become excessive, so firms stop importing and lower markups to sell down their stock.

8.3 Incremental contribution revisited

The paper contributes by linking three literatures: trade costs, inventory theory, and exchange-rate pass-through. It shows that the microstructure of trade costs matters for both welfare calculations and macroeconomic adjustment. In particular, the same friction that explains lumpy micro trade can also explain aggregate devaluation dynamics.

8.4 Implications for trade and macroeconomics

- Trade policy and infrastructure that reduce delays or fixed transaction costs can have effects much larger than direct cost measures suggest.
- Emerging-market trade collapses may partly reflect inventory adjustment, not only financial constraints or demand collapse.
- Exchange-rate pass-through can be gradual even without sticky prices because inventories affect the marginal valuation of goods.
- Extensive-margin trade responses can be very large in the short run even when the set of potential goods and firms is unchanged.

8.5 Limitations

- The model is partial equilibrium and takes devaluation shocks as exogenous.
- Inventory positions are not directly observed in all devaluation datasets, so recent shipments and lumpiness are used as proxies.
- The model abstracts from financing constraints, firm balance sheets, and general-equilibrium feedbacks that may also matter in emerging-market crises.
- The demand-shock volatility needed to match inventories should be interpreted broadly, not literally, because it captures omitted uncertainty as well.

8.6 Takeaway

The paper’s main lesson is that trade is not frictionless flow through an iceberg. It is a sequence of costly, delayed, and lumpy orders managed through inventories. Once trade is understood this way, the large short-run collapse of imports and slow retail price pass-through after devaluations become natural outcomes of firms’ inventory decisions.